HMC ONE
AI STRATEGY. BUILT TO SCALE.

CLIENT CASE STUDY · WIRES & CABLES MANUFACTURING · PROCUREMENT

33,000+ SKUS · HOUSE WIRES, POWER, CONTROL & SPECIALTY CABLES

Engineered BOMs

A calibrated BOM engine cut new-SKU BOM time from 1.5 days to 30 minutes and took the guesswork out of copper and compound buys.

~~1.5 days~~ → **30 min** BOM turnaround for a new SKU



30 min
BOM turnaround, new SKU
▼ 97% vs 1.5 days



98.6%
BOM accuracy vs actual consumption
▲ +5.5 pts vs 93.1%



1.0%
Conductor buffer built into BOM
▼ from 3.5%

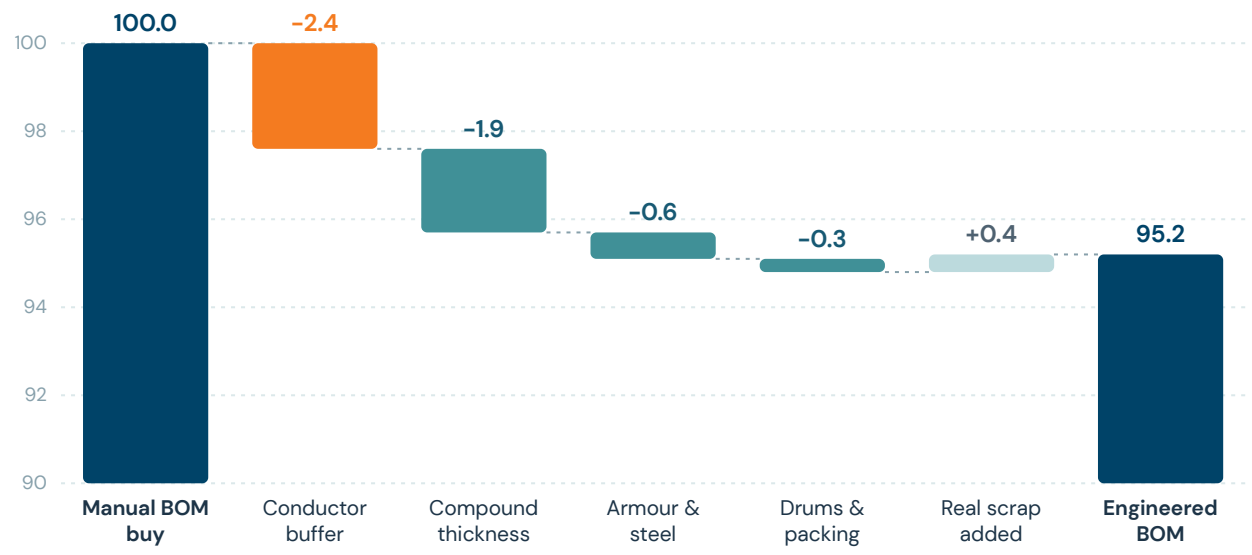


4.8%
Material buy avoided
▲ vs manual-BOM baseline

Rings: share of BOM time removed; accuracy with the earlier 93.1% inside; buffer as a share of the old 3.5% (inner ring = old level); saving levers, conductor buffer in white. Figures indicative; validated metrics available on request.

01 Where the material saving came from

Bridge from manual-BOM buy to engineered-BOM buy · indicative



₹ per ₹100 of material bought on manual BOMs · axis starts at 90

BIGGEST LEVER
Stacked margins, not physics
Conductor buffer (-2.4) and compound thickness (-1.9) were most of the saving: upper-tolerance thickness plus a flat margin at every step.

HONEST ALLOWANCE
Real scrap added back: +0.4
Process allowance now comes from actual scrap by plant and material, not a flat margin; it adds +0.4 per ₹100 back.

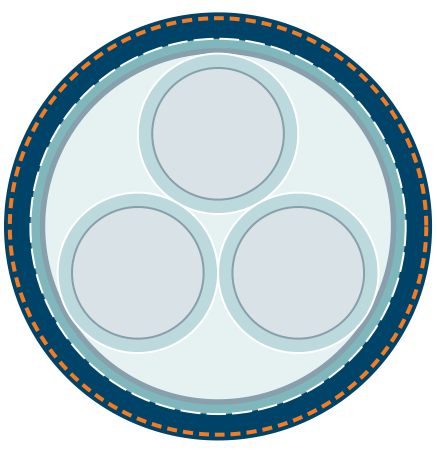
SMALLER LEVERS
Armour, steel, drums, packing
-0.6 per ₹100 from armour & steel and -0.3 from drums & packing.

NET RESULT
₹100 → ₹95.2 per ₹100 bought
4.8% material buy avoided against the manual-BOM baseline.

02 Every layer of the cable, priced per kilometre

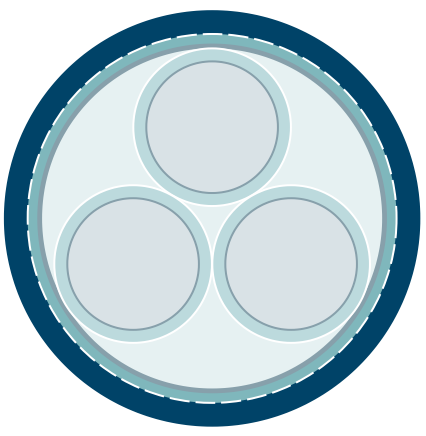
The engine's illustrative model · LV power construction

MANUAL BOM



Upper-tolerance thickness
Ø 35.8 mm; orange ring = engineered outline

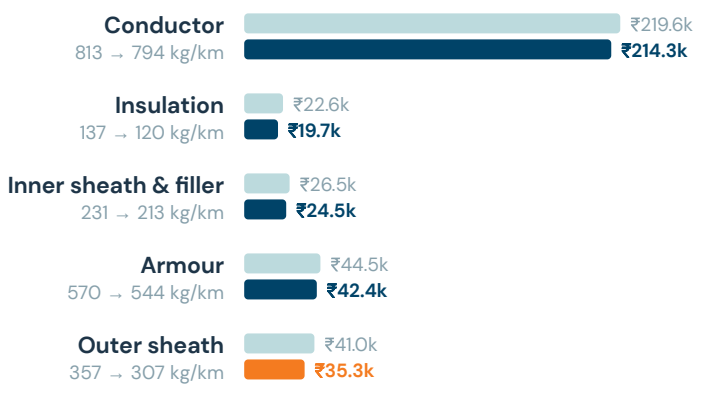
ENGINEERED BOM



Nominal geometry
Ø 34.8 mm from the parameter master

Aluminium conductor · XLPE insulation · Inner sheath & filler · Strip armour · Outer sheath

3C × 95 sq mm Aluminium · XLPE insulated · strip armoured
LV power cable · IS 7098-1 construction · material cost per km, manual (light) vs engineered (dark)



MATERIAL PER KM
~~2,108~~ → **1,977 kg**
▼ 6.2% vs manual BOM

MATERIAL COST PER KM
~~₹354k~~ → **₹336k**
▼ 5.1% · saves ₹18,040/km

CONDUCTOR BUFFER
~~3.5%~~ → **1.0%**
Set from actual scrap data

How it works: layer geometry from nominal thickness and lay-up factors · mass from volume × density, layer by layer · process allowance from actual scrap, not a flat margin.

Material prices and allowances indicative. Construction parameters simplified for illustration.

03 Four pain points, four fixes

Before → after, for procurement

BEFORE
A spreadsheet per SKU
Engineers recalculated kg/km by hand for every new SKU or tender: one to two days per BOM.

PYTHON ENGINE · EXCEL FRONT-END

AFTER
BOM in ~30 minutes
Generated from the construction; Claude extracts tender-spec parameters into the template for review.

BEFORE
Buffers on buffers
Upper-tolerance thickness plus a flat margin at each step; copper and PVC over-bought and held as stock.

PHYSICS APART FROM POLICY

AFTER
Allowance from actual scrap
Nominal geometry from a governed parameter master; allowances by plant and material, owned by procurement.

BEFORE
One cable, many BOMs
The same cable code carried different BOMs across plants and files; none matched the approved construction.

VERSIONING · SIGN-OFF

AFTER
One versioned BOM
A single BOM per cable code with an engineering–procurement sign-off workflow.

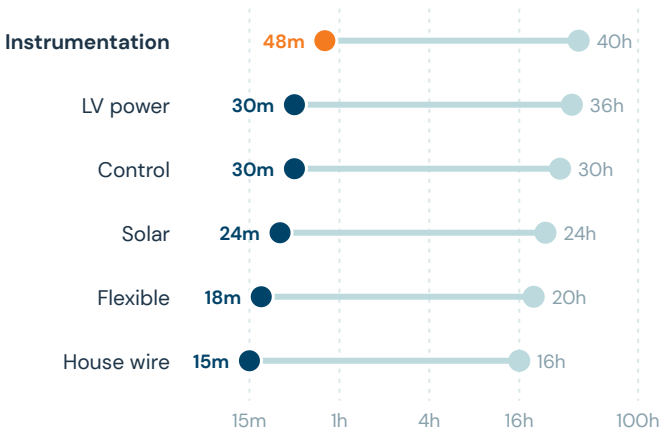
BEFORE
MRP on stale numbers
Plans missed spec changes and real scrap rates, causing compound shortages and spot buys.

MONTHLY RECALIBRATION

AFTER
ERP-ready, kept current
ERP upload sheet from the engine; allowances recalibrated monthly against consumption.

04 Days became minutes

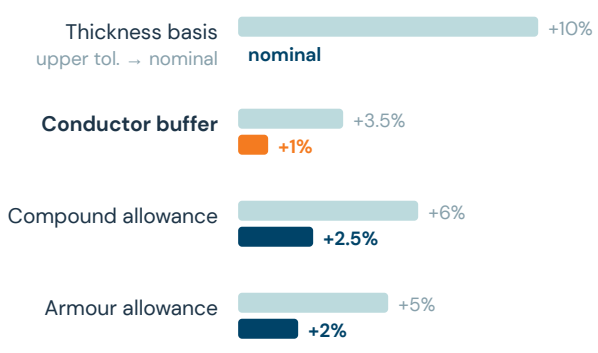
Hours per new-SKU BOM by product family · indicative



Log scale · light = manual, dark = engineered

05 The buffers, made visible

Allowance policy, manual (light) vs engineered (dark)



3.5% → 1.0% conductor buffer in the BOM

06 How HMC ran it

3 weeks	**Discover** 200 high-volume SKUs checked against issue slips and drum weights; variance baseline by family.
2 weeks	**Design** Construction logic aligned to IS specs; parameter master and allowance policy table.
4 weeks	**Build** Python engine, Excel/VBA front-end, Claude spec extraction, Power BI variance view.
4 weeks	**Adopt** LV power pilot, six-week parallel run; MRP switched once variance held below 1.5%.
∞ ongoing	**Scale** Every family and plant; monthly recalibration; a named owner for the parameter master.

What transfers to any engineered-product buyer

01 — Separate physics from policy
Geometry and density are fixed. Buffers are a business decision. Keep them in separate tables so each can be challenged.

02 — Calibrate against consumption
Use actual issue data to set allowances. Opinions about scrap are usually higher than scrap.

03 — Give the master data an owner
One accountable owner for the parameter master kept 33,000 SKUs consistent long after go-live.

Built with **Claude** · **Python** · **Excel / VBA** · **Power Query** · **Power BI** · **ERP upload template** | Prepared by Havish M Consulting · Figures indicative; validated metrics available on request.

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